

Aerial Robotics

FIELD AND SERVICE ROBOTICS

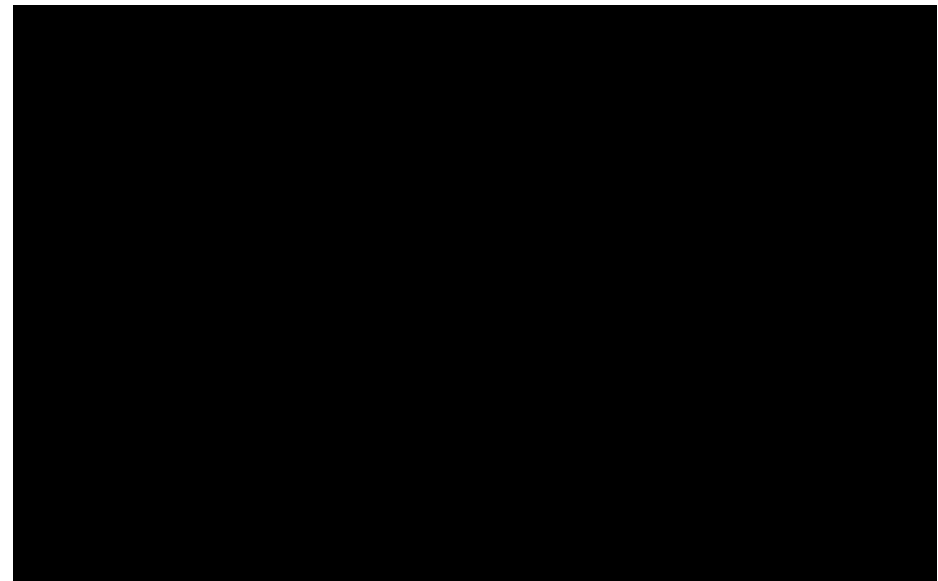
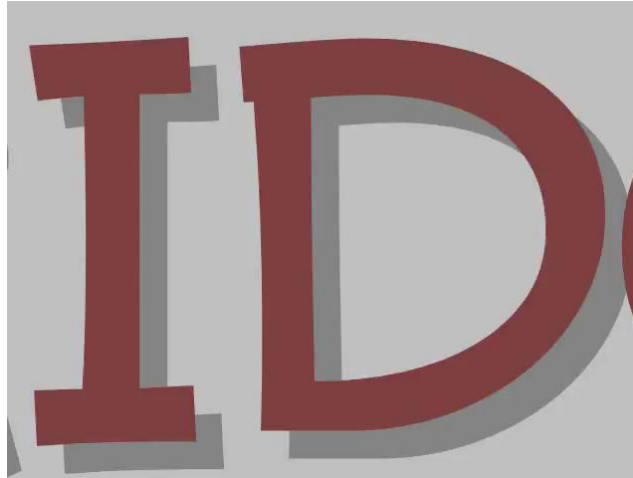


www.prisma.unina.it

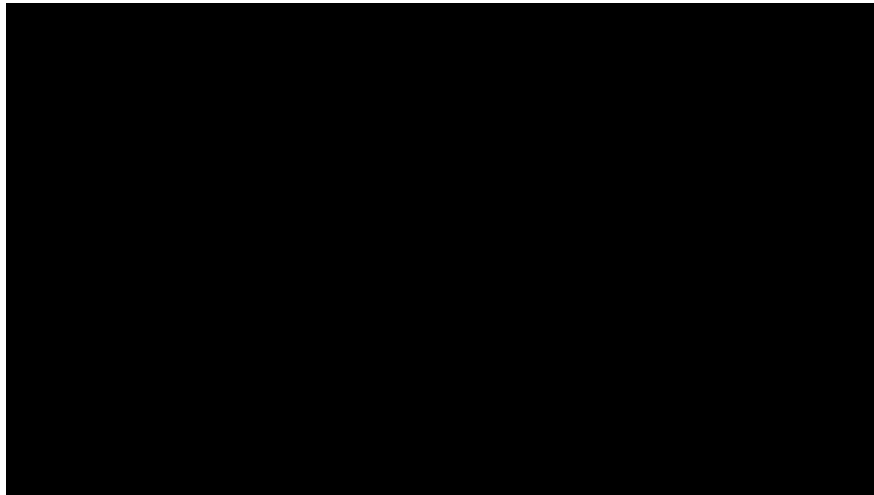
- Aerial robotics has become a new frontier in field and service robotics



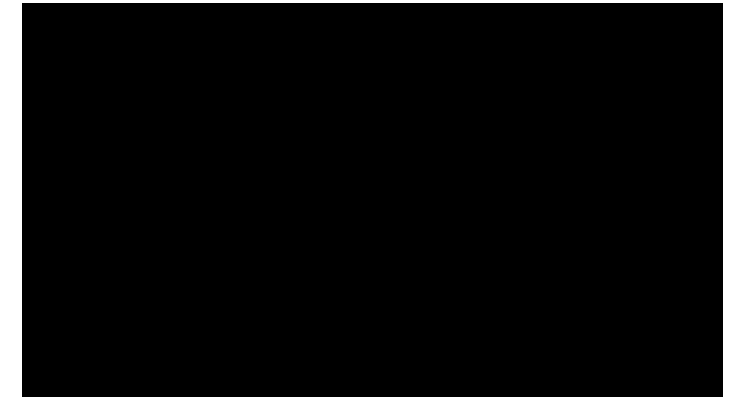
- Really useful?



■ Business opportunity



Delivering?



Industry?



Marketing?



■ Aerial robotics applications

- Transportation
- Manipulation
- Inspection and maintenance (I&M)
- Repairing
- Monitoring
- Patrolling
- Exploration
- Intervention in devastated areas
- ...

■ Oil and gas facilities

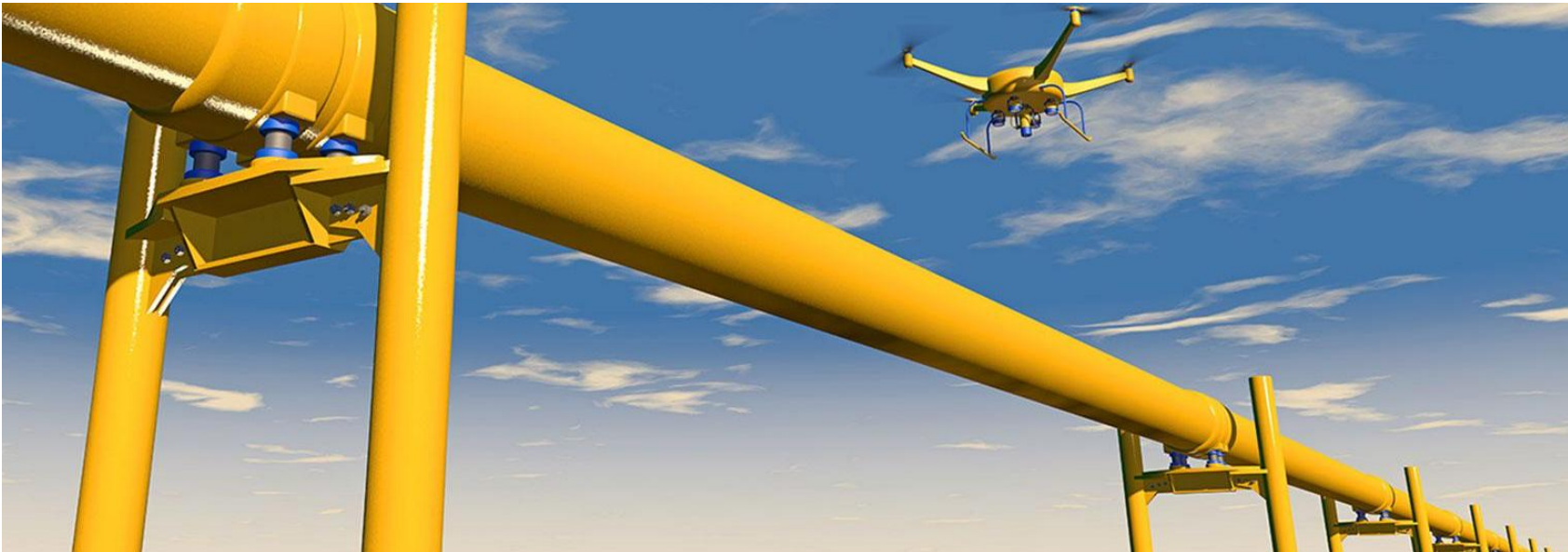
- What is possible:
 - 3D mapping of existing plants
 - Visual inspection
 - Thermographic inspection
 - Direct measurements (e.g., thickness)
 - First intervention and monitoring
- What is needed:
 - ATEX compliant solutions
 - Power remote-handling for maintenance and (dis)assembling



- Drone for inspection in confined spaces



- Drone for long pipe inspection



■ Drone for thickness measurements



History

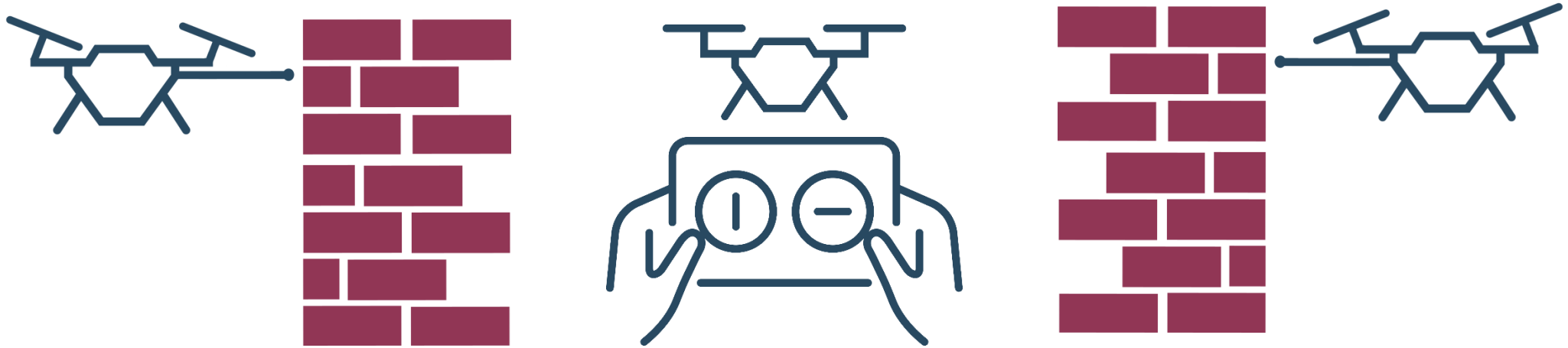
- Leonard Da Vinci's notes
- First autonomous flight of an **unmanned aerial vehicle** (UAV) in 1896
 - Number 5 vehicle, designed by Samuel P. Langley, that travelled 400 meters
 - Number 6 vehicle travelled 1600 meters
- In 1916, during the First World War, the Aerial Target vehicle was radio controlled (RC)
- Hewitt-Sperry vehicle in 1916



- Before the Second World War the airplanes were converted into autonomous vehicles controlled by autopilots
 - The first society, the *Radioplane Company*, specialized in building was founded by Reginald Denny
 - 1500 radio-controlled helicopters were built for the Second World War
- Technological development, smaller processors, lightweight sensor measurement systems, global navigations systems and so on boosted UAVs growth
 - In Japan, Yamaha Motor company developed an unmanned helicopter for irrigation in 1983
 - An autonomous UAV flying with GPS was out in 1988
 - It observed volcanic activities at Usu-zan in 2000

How is a drone made?

Getting started with the hardware



■ “Baby-Koala” Co-axial Tilting Octa-Rotor

■ Hardware

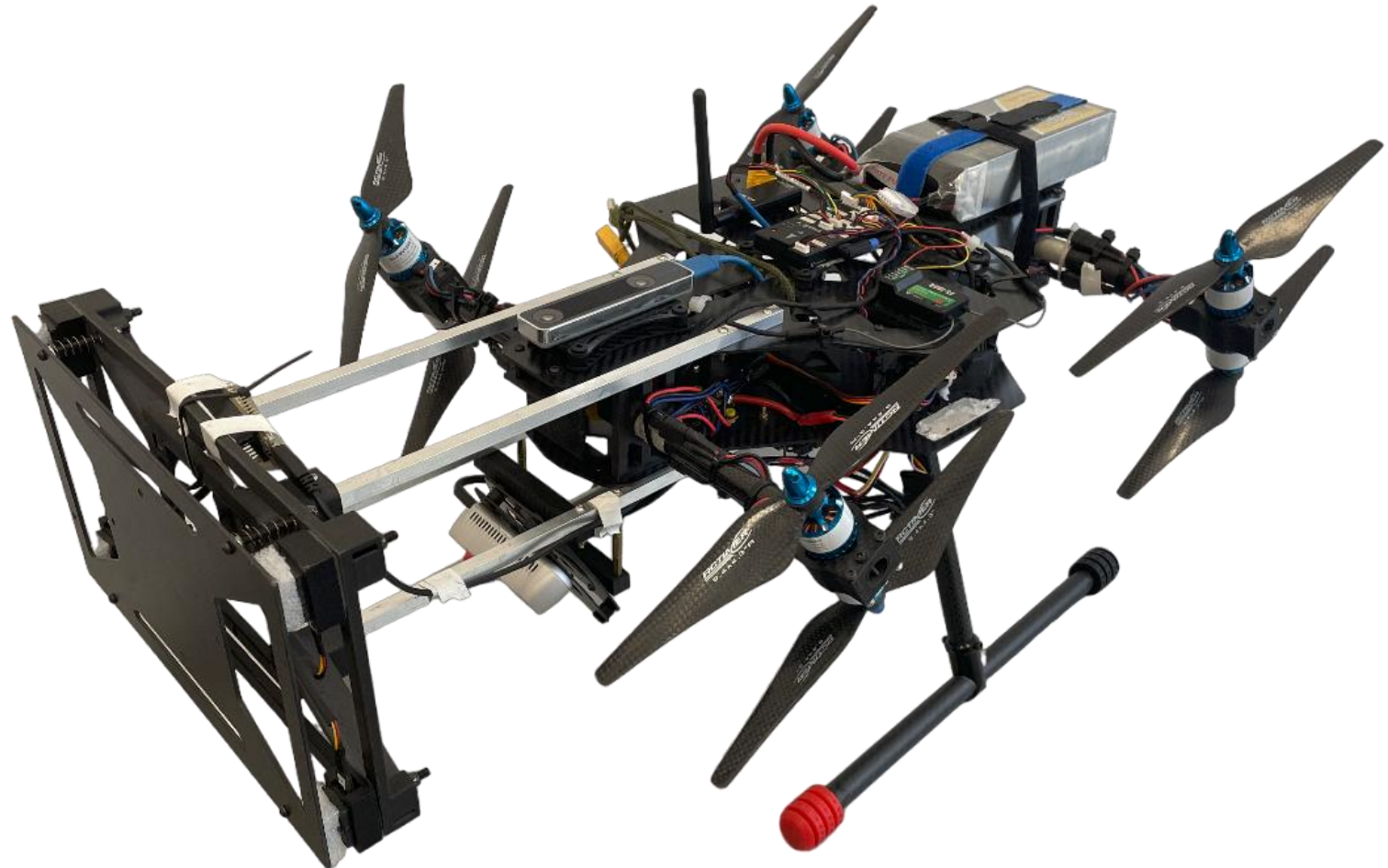
- Aluminium & Carbon fibre frame
- Propellers
- Motors
- ESC
- LI-PO Batteries

■ Perception & Control

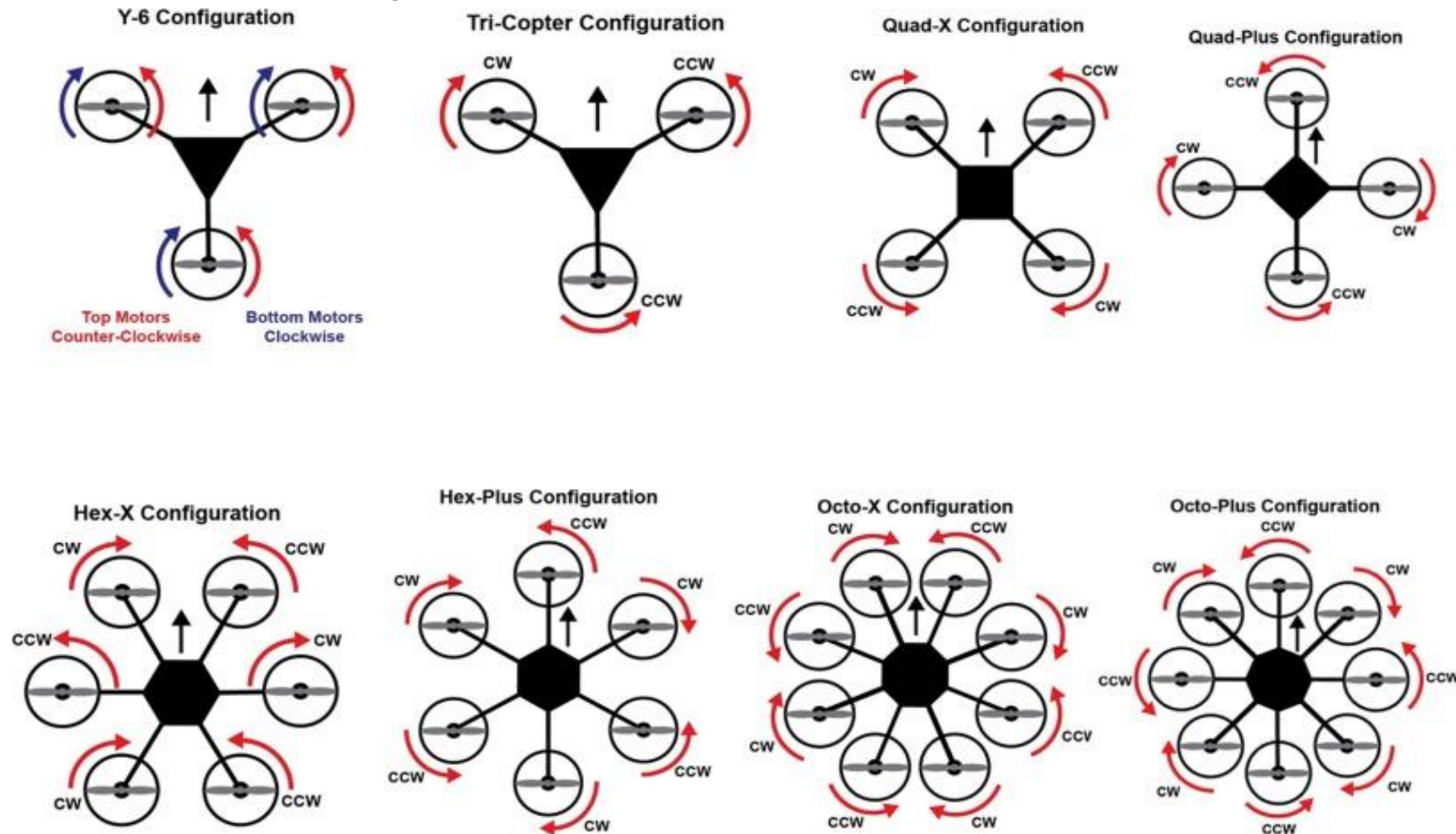
- Flight Controller - Pixhawk Cube Orange
- Onboard Computer
- Sensors
- Tools

■ Communication

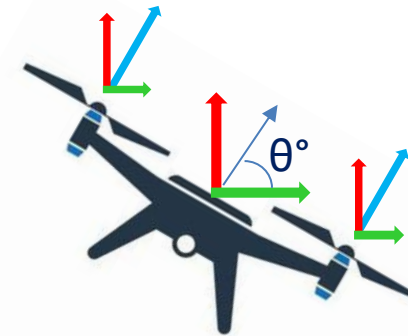
- Telemetry, RC receiver, WI-FI Antenna



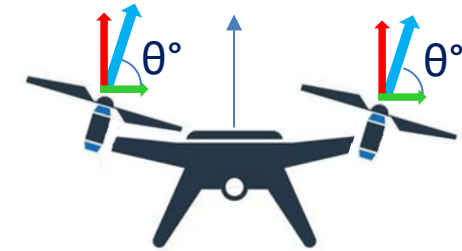
■ Frame – several multirotor configurations



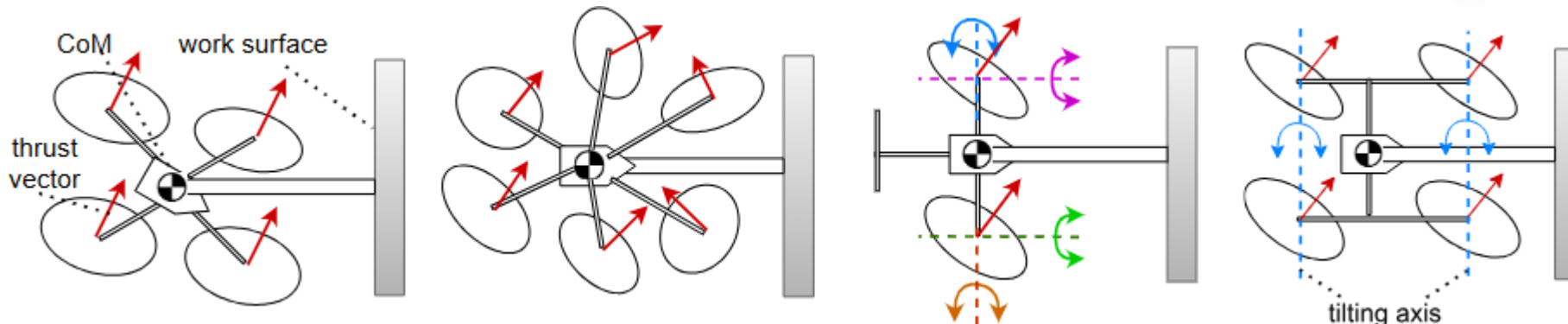
- Conventional multi-copters need to change their orientation to move in the horizontal plane
 - 4 controllable DoFs
 - During hovering, roll and pitch are indirectly controlled to maintain the hovering stability
- While drones with tilted (fixed angle) or tiltable/tilting (time-varying angle) can have a desired orientation while moving
 - Tilting drone with independent servo-motors: 6 controllable DoFs and possibility to impose a desired roll or pitch angle



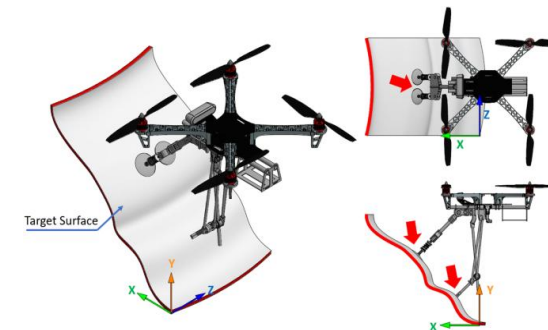
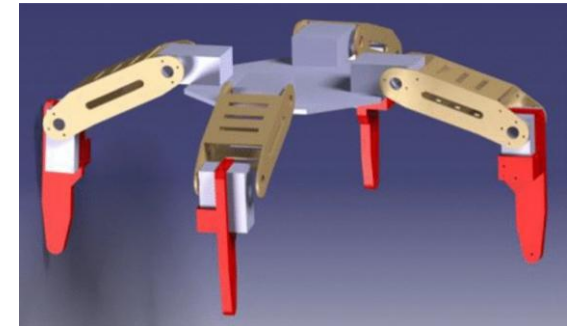
Standard UAV



Tilting UAV



■ Landing gear



■ Characteristic

- **Length** of a propeller (its diameter): larger propellers need more power and create a bigger pressure difference. They generate more thrust
- **Pitch** Defined as the travel distance of the propeller per revolution, generally measured in *inches*
- Number of **blades**
- Spin: **CW** and **CCW**

■ Motors

- DC Brushed
- DC Brushless
 - Need **ESC** (Electronic Speed Controller)
 - Controlled with **PWM** signals (converting a velocity command)
 - Basic parameters are:
 - Maximum voltage/ Maximum LiPo cells
 - **Kv**: number of revolutions per minute (rpm) that a motor turns when 1V (one volt) is applied with no load attached to that motor



Batteries

- How to choose? Flight time, weight, load etc → capacity, current
- Knowing the “Kv” rating of a motor will help you determine how fast that motor will rotate when a given voltage is applied to it.
 - Ex: a 980Kv motor powered by an 11.1V battery would spin at 10,878 rpm (980×11.1) with no load



Li-Po



Li-ion

■ Pixhawk flight controllers

The flight controller is the board where the brain and the main senses of the drone are placed.

Pixhawk

Originally manufactured by 3DR.
Discontinued in 2016.



Pixhawk 2

Originally manufactured by 3DR for Solo.
Discontinued in 2016.

Alternative: HEX CubePilot Black



Pixhawk 3 Pro

Manufacture: Drotek
Architecture: FMUv4



Pixracer

Manufacture: mRobotics
Architecture: FMUv4

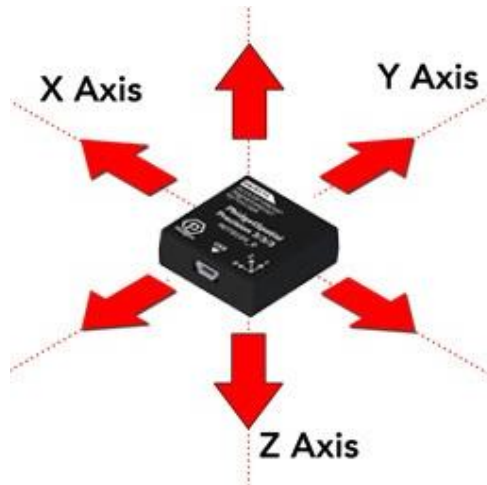


Pixhawk 6C & 6x

■ Sensors

There are some fundamental built-in sensors to allow the drone to stabilize and navigate

Accelerometer



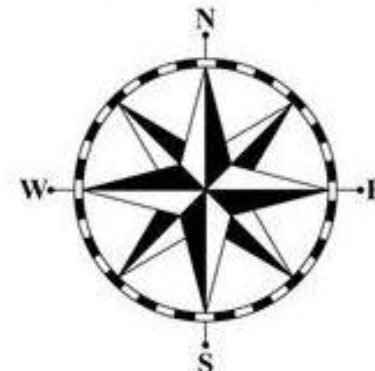
Acceleration
along 3 axes

Gyroscope



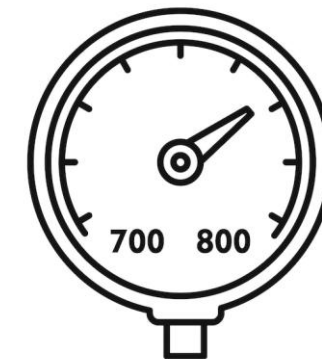
Rotation speed
Around 3 axes

Magnetometer



Direction of
magnetic north

Barometer



Altitude

■ Vision sensors

- RGB camera
- RGB-D camera



■ 3D Lidar

- Based on time-of-flight scans (from 16 to 32 scan lines)
- Based on rotating mirror hybrid-solid technology



■ Optical flow/Range sensors

- downward facing camera and distance sensor for velocity estimation. It can be used to determine speed when navigating without GNSS



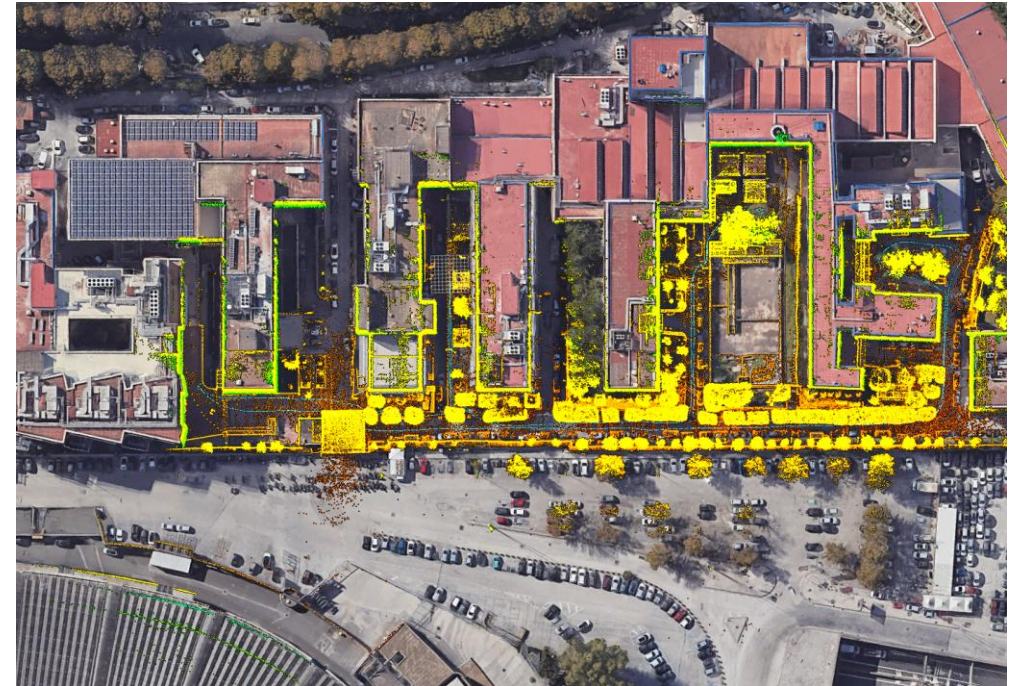
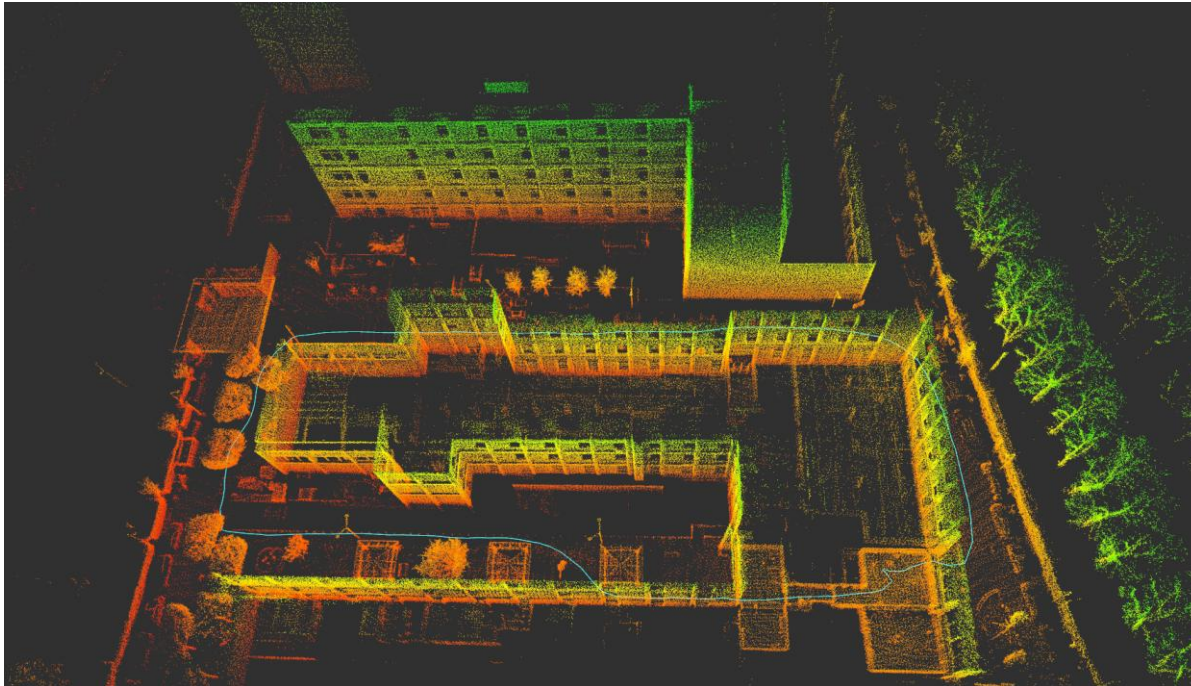
■ GPS

- Needed in outdoor environments with poor features



3D Lidar

- Laser scan computing the distance from obstacles
- point cloud to reconstruct the environment



■ Unmanned Aerial Manipulators (UAM)

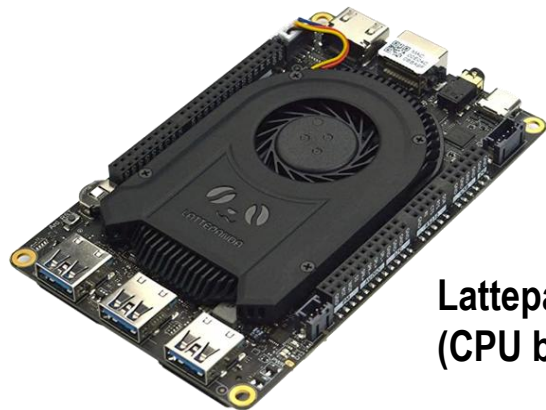
- Aerial platform equipped with rigid sticks or articulated manipulators
- Usually used in contact-based NDT tests
- Interaction force has to be controlled to avoid damages

■ Force sensors

- Load cells
 - Mono-dimensional sensors
 - Retrieve the force measure converting the material deformation in electric signal
 - Cheap and fragile
- ATI 6-DOF sensors
 - Compact and rugged sensor for measuring forces and torques on six axes
 - Components: Includes transducer, shielded high-flex cable, and data acquisition system
 - High costs



- **Onboard computer** must be chosen based on the computing power and depending on the application.



LattePanda 3 Delta
(CPU based)



NVIDIA Jetson
(GPU based)

- **Communication links**

There are several ways to communicate with a drone



**RC &
Radio receivers**

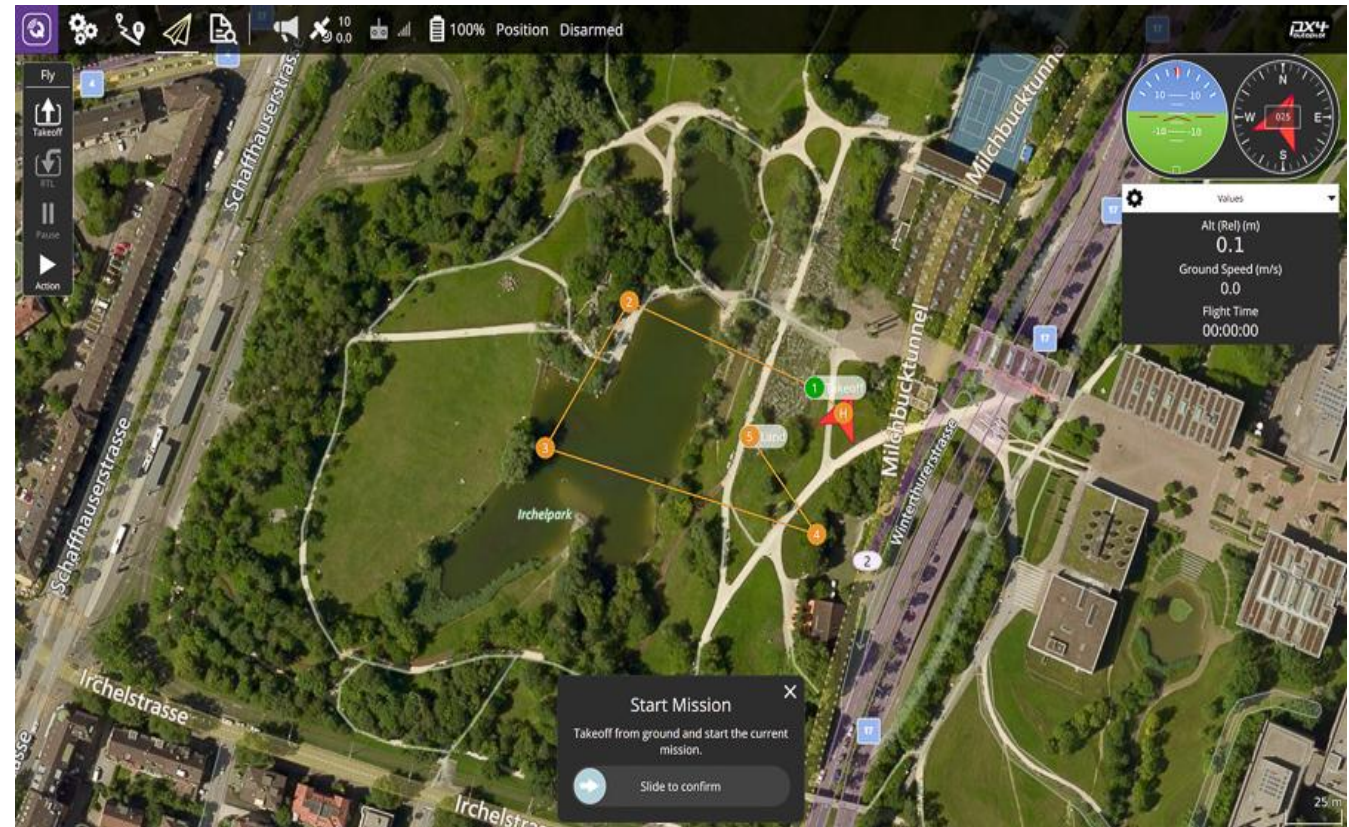
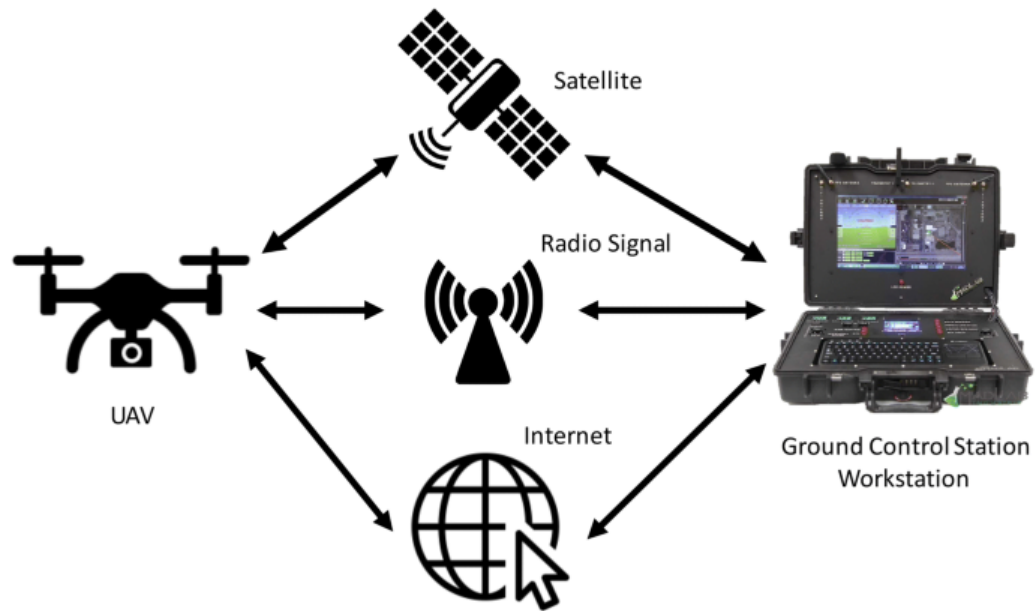


Radio telemetry



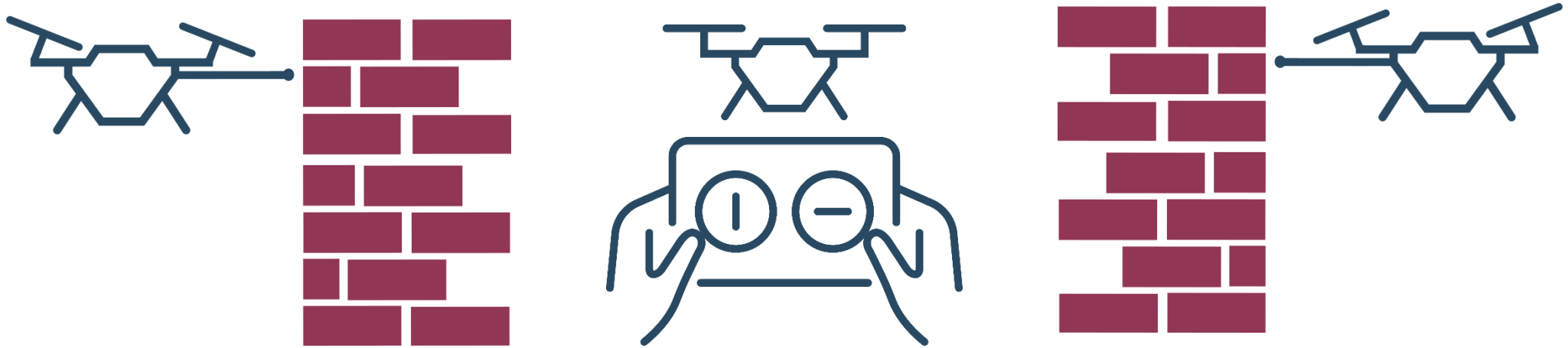
Wi-Fi antenna

- The ground station is the control centre that provides the facilities for human control of the drone
"It is the unit that doesn't fly" and gives supports to the drone



How can we program a UAV?

Getting started with software



■ Firmware/Autopilot

The firmware is the set of programs that allow the drone to fly, navigate and communicate

- The autopilot is the "brain" of the drone.
- It minimally consists of *flight stack* software running on a real time OS ("RTOS") on *flight controller* (FC) hardware.
- The flight stack provides essential stabilisation and safety features, and usually also some level of pilot assistance for manual flight and automating common tasks, such as taking off, landing, and executing predefined missions.



■ Main features of the **PX4** Autopilot:

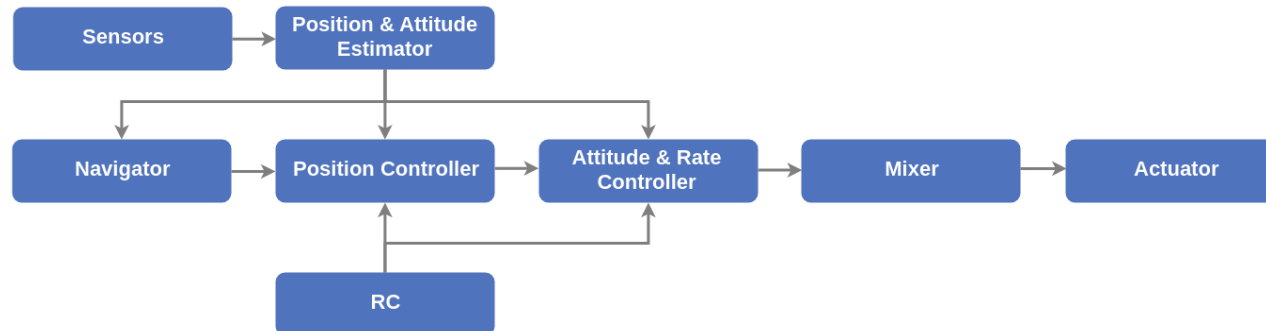
- Open-source
- Modular Architecture
- Integrated with ROS/ROS2
- Multiple vehicle support
- Fully manual, partially assisted and fully autonomous flight modes
- Vehicle stabilization
- Waypoint navigation
- Sensor fusion (EKF)
- Compatible with the QGroundControl ground control station software



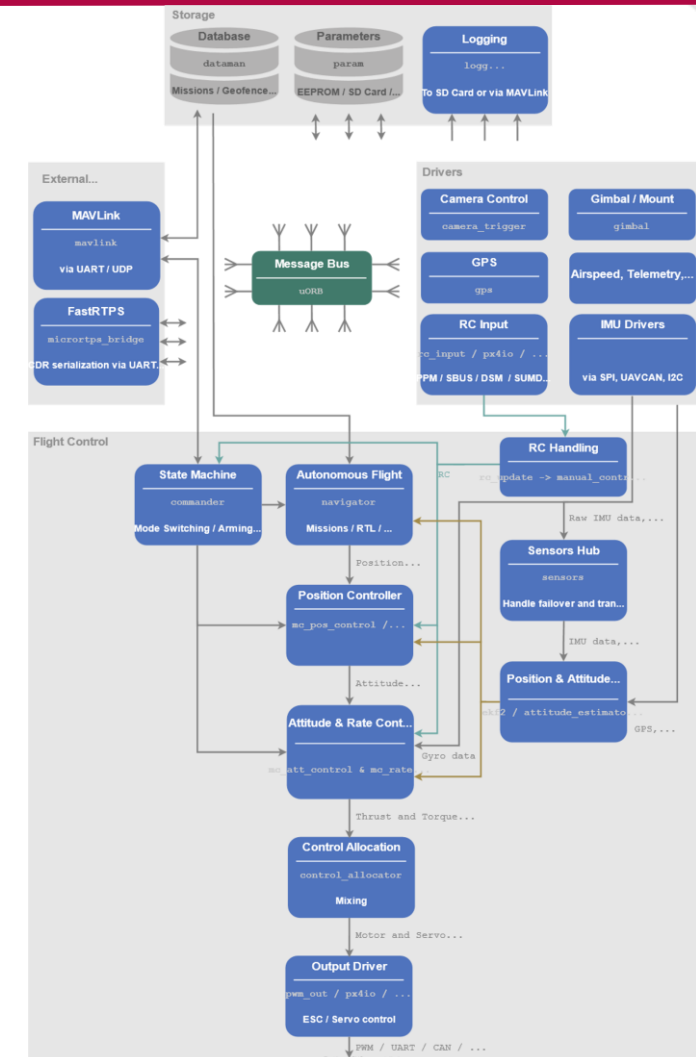
■ PX4 consists of two main layers:

- the **flight stack** is an estimation and flight control system,
- and the **middleware** is a general robotics layer that can support any type of autonomous robot, providing internal/external communications and hardware integration.
- Modules communicate with each other through a publish-subscribe message bus named **uORB**.

■ Flight Control Stack

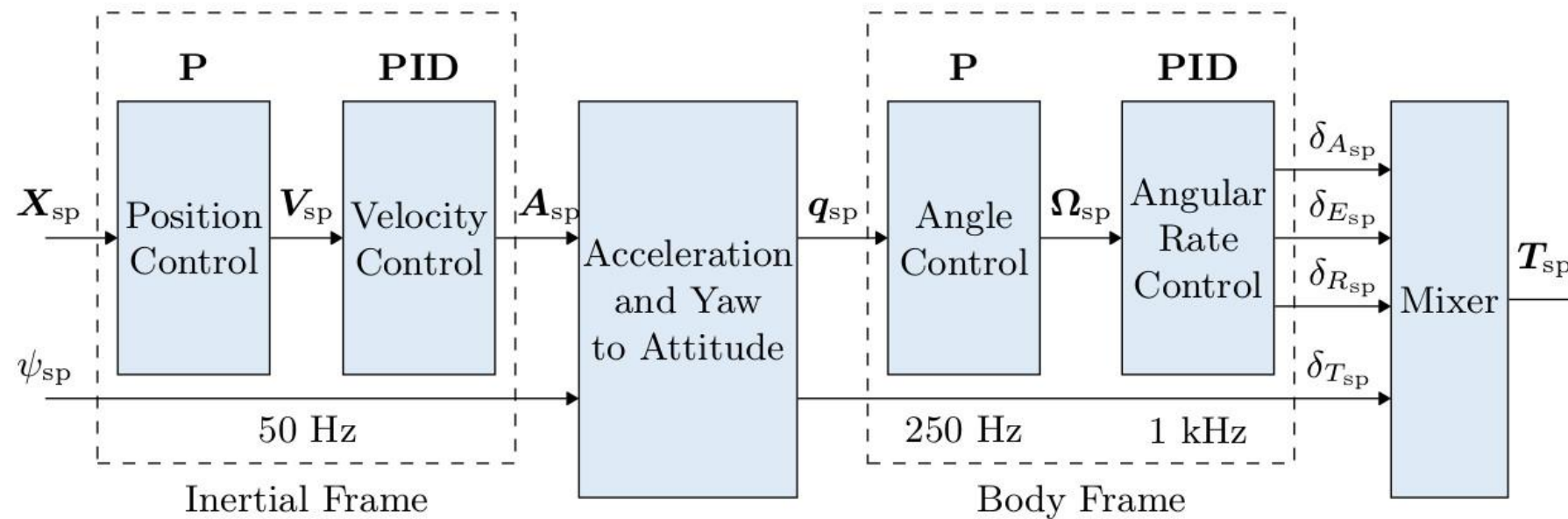


(<https://docs.px4.io/master/en/concept/architecture.html>)



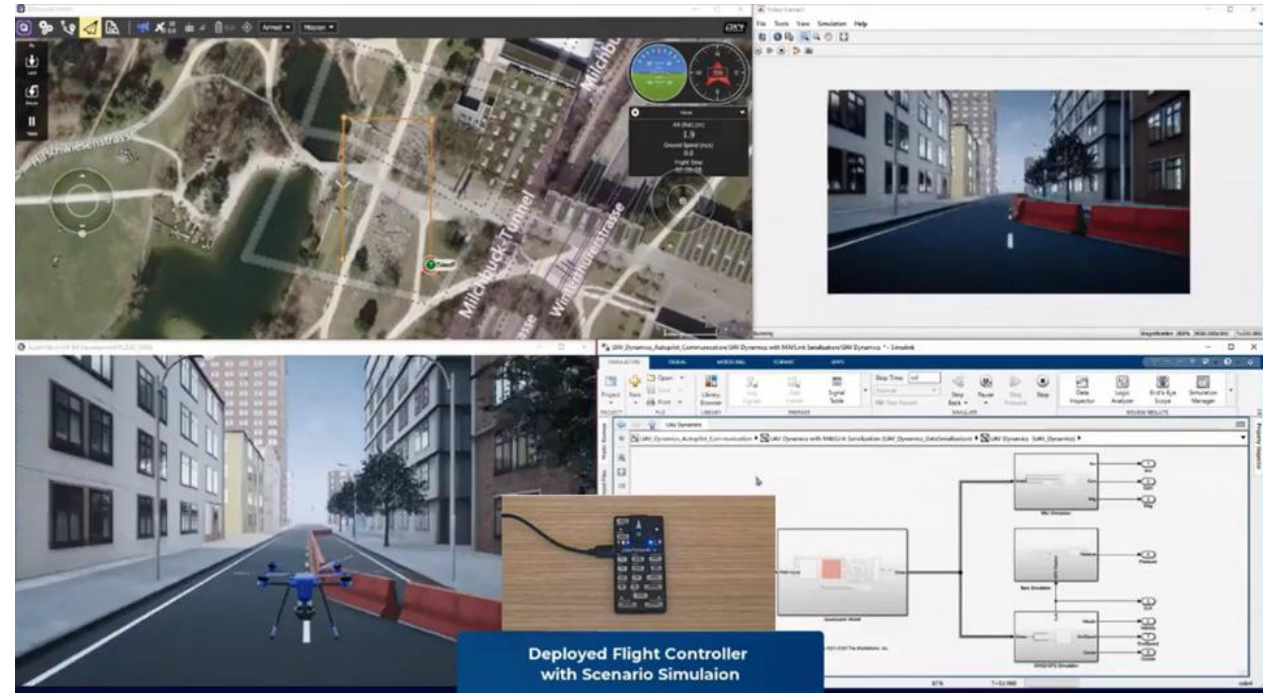
High-Level Software Architecture

- PX4 controller diagram (https://docs.px4.io/master/en/flight_stack/controller_diagrams.html)



- It is possible to interact with the firmware at different levels
- **High level**
 - The firmware remain the same, we develop solutions for
 - Path and Trajectory planning
 - Localization and mapping
 - Control equipment in a decentralized way (manipulator, gimbal, etc.)
- **Low level**
 - The firmware needs to be modified to
 - Implement a custom controller requiring low computational power
 - Modify the internal EKF logic
 - Add firmware-level functionalities
 - Add support for new airframes

- <https://it.mathworks.com/help/uav/px4-spkg.html>
- We can implement our flight control and/or mission planner in Simulink, test it and then upload directly on the main board (flight controller).
- Advantages
 - Easy and fast prototyping
 - Use of MATLAB features
 - Easy Environment Set-up
 - Windows support
 - C code generation
- Some drawbacks of this approach:
 - High Pixhawk CPU usage
 - Not working with latest PX4 Firmware releases
 - Cross compiled code: unreadable
 - Loss of some features

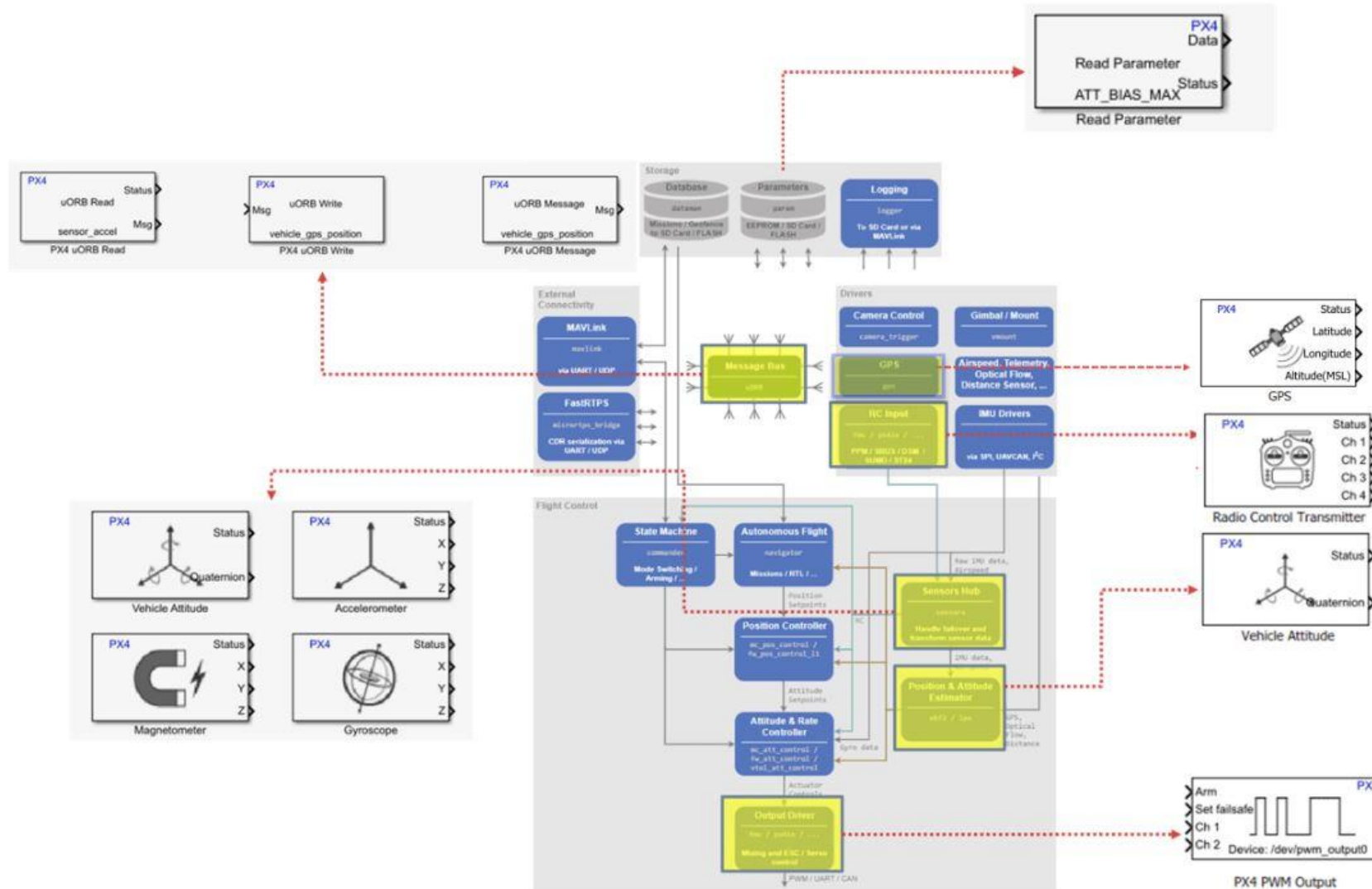


MATLAB®
& SIMULINK®

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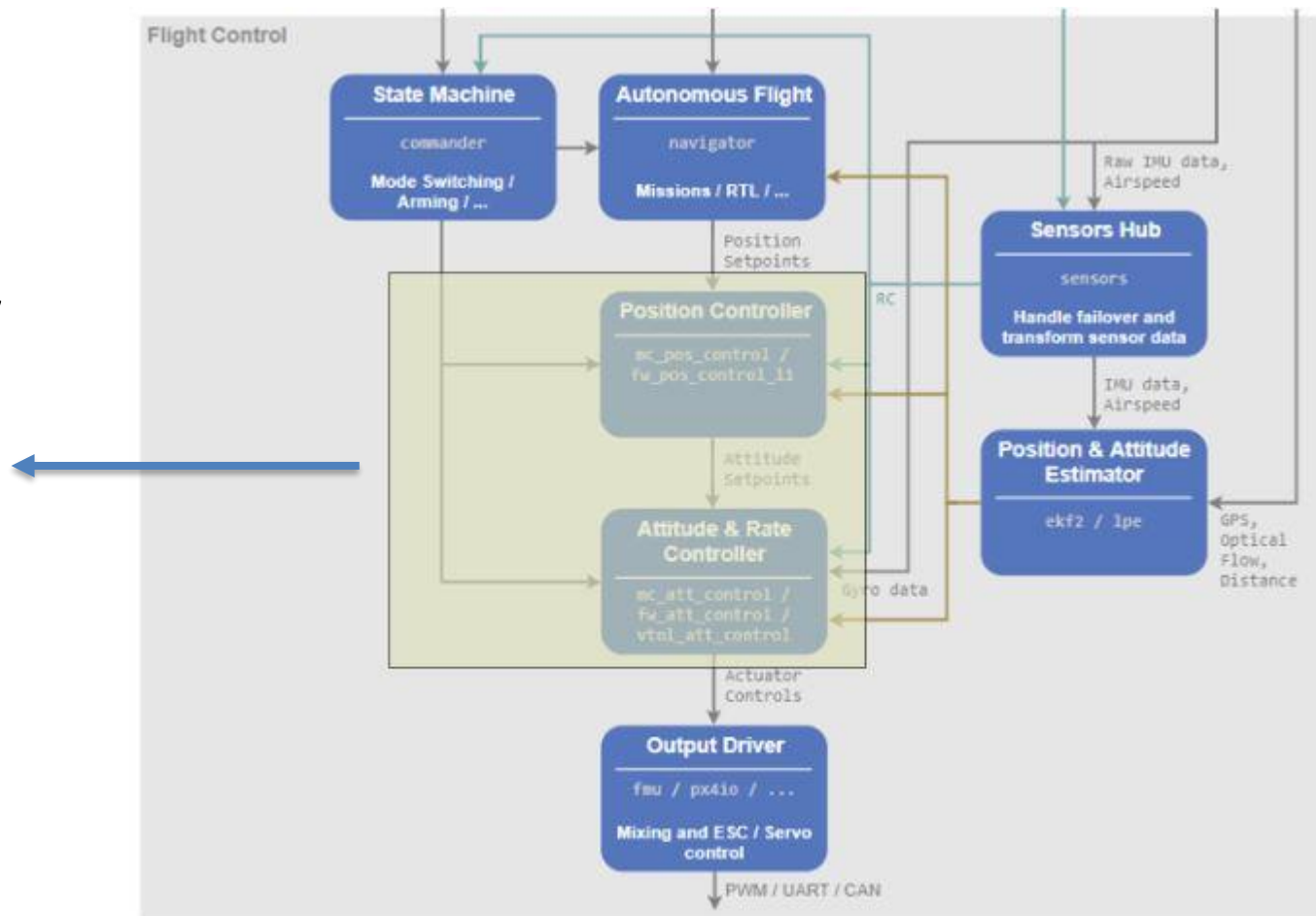


Toolbox blocks

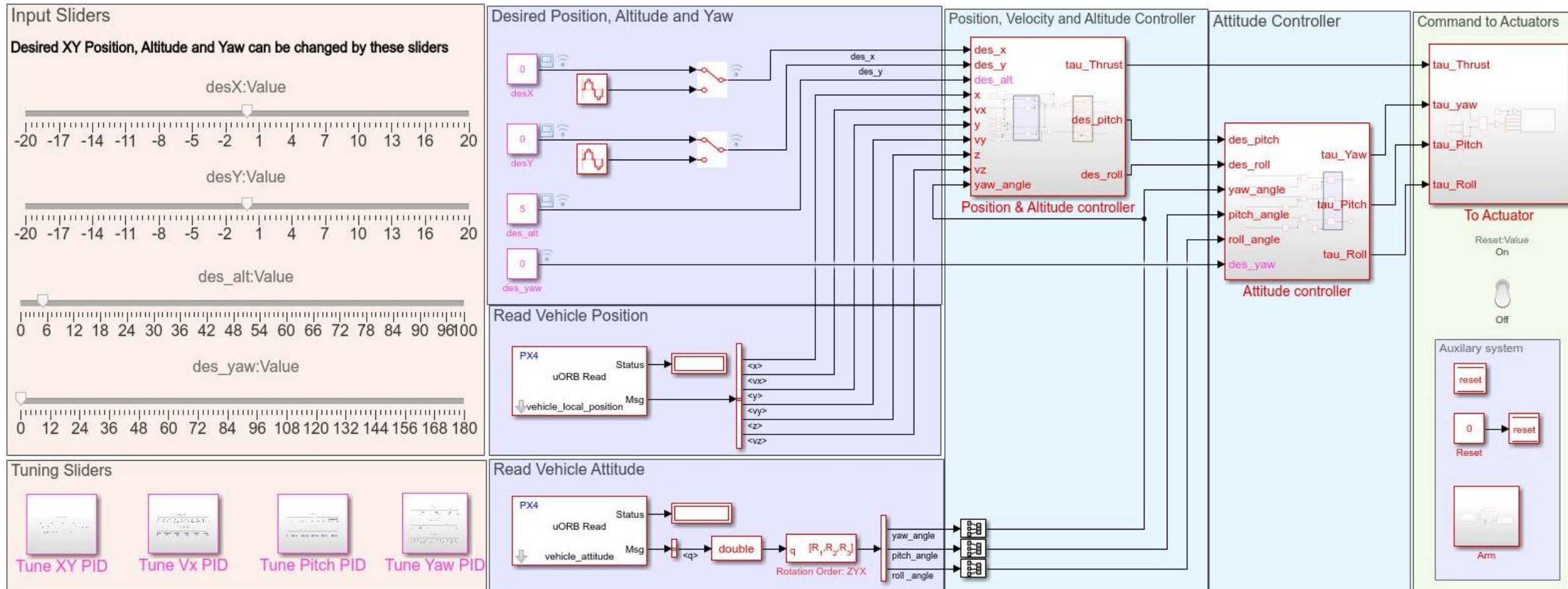


- We can change the px4 flight control with a custom one

- Geometric tracking controller
- Impedance Control
- Motion/force Hybrid control
- and so on...



Example of a position control with the PX4 Simulink toolbox



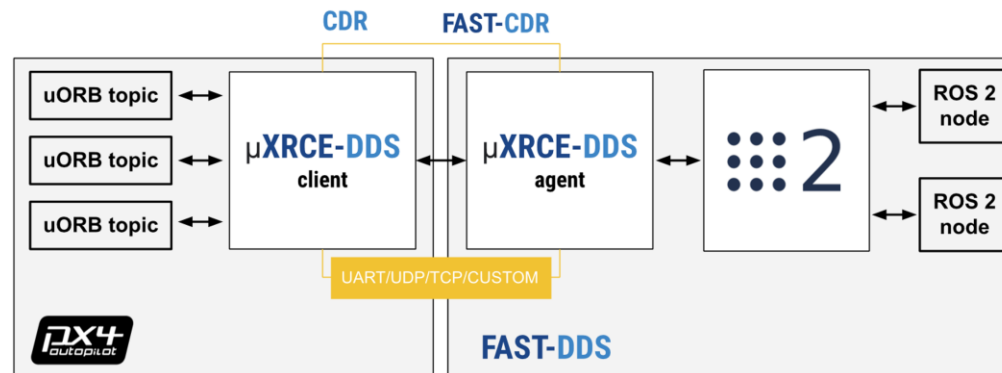
- An initial autonomous mission completed with the customized PX4 firmware



More complex algorithms (e.g. vision based, SLAM) can be executed on the onboard computer and the communication can be handled with ROS

■ ROS/ROS2 (https://docs.px4.io/main/en/ros2/user_guide)

- ROS 2: PX4 and ROS 2 communicate over the *PX4-ROS 2 bridge*, an interface that provides a direct bridge between PX4 *uORB* messages and *ROS 2 DDS* messages/types. This effectively allows direct access to PX4 internals from ROS 2 workflows and nodes in real-time.



ROS2

+



- ROS 1 via MAVROS: deprecated

■ First flight

- Clone on your machine the [aerial robotics](https://github.com/PX4/PX4-Autopilot) repo
- Build the Docker image (ROS2 Humble, GZ Harmonic and PX4 dependencies)

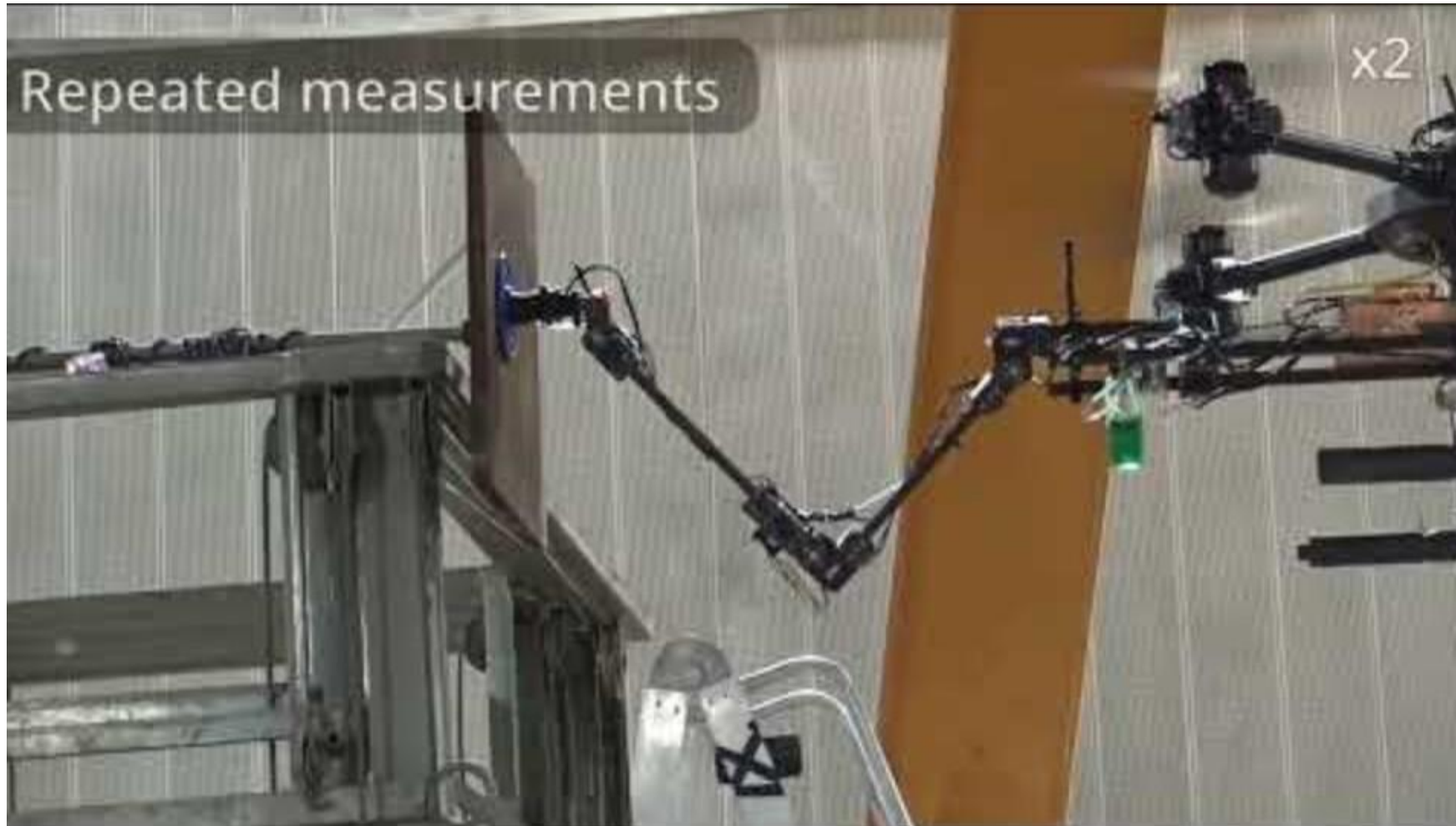
```
$ git clone https://github.com/PX4/PX4-Autopilot.git --recursive
$ cd PX4-Autopilot
$ checkout v1.16.0
$ make submodulesclean
$ make px4_sitl gz_x500
```

- Download and install [QGroundControl](#)
- Take-off and flight!

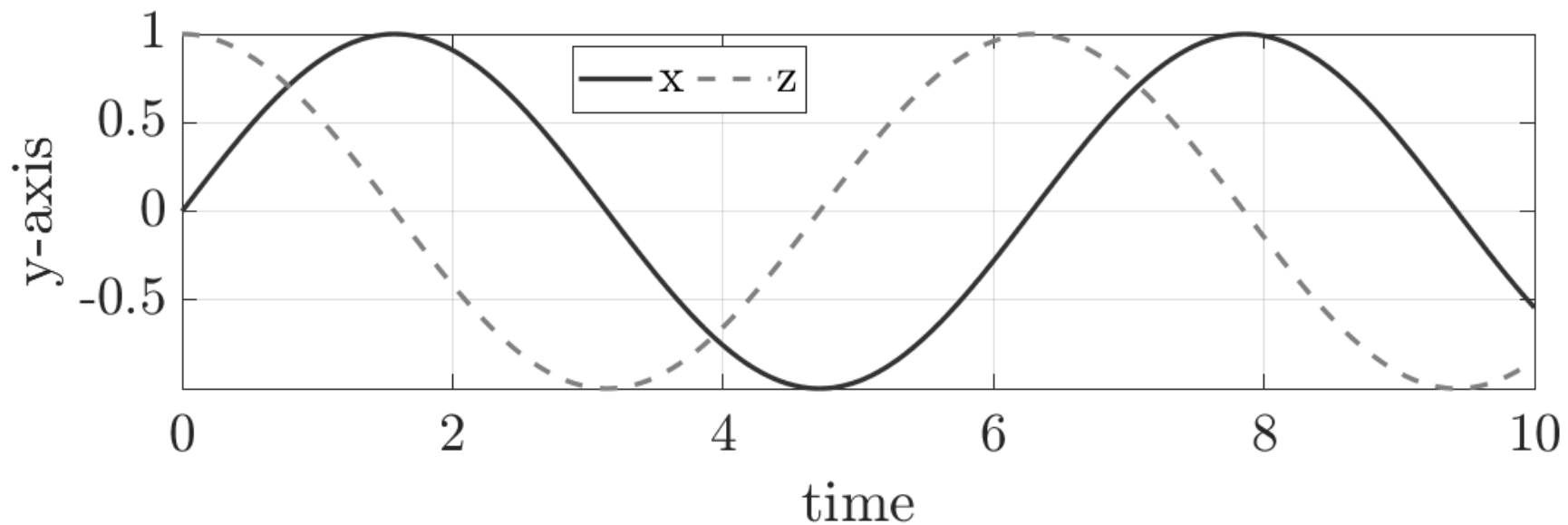
- px4 msgs are needed to be sourced in your workspace
- Launch GZ simulation
- Enable communication through uxrce_dds:
 - `$ MicroXRCEAgent udp4 -p 8888`
- Launch a dedicated ROS2 controller
 - Useful examples:
 - https://github.com/SaxionMechatronics/px4_offboard_lowlevel
 - https://github.com/Simone-DAngelo/geometric_tracking_controller
- Gain tuning and parameters are airframe-dependent
- The message OffboardControlMode specify the expected input to the px4 framework bypassing built-in controller

- **PX4 control stack is model-free**
- No information is needed in terms of Inertia, Mass, etc.
- The geometry must be known
 - Airframe
 - Motor positions
 - Servo motors limits
- The thrust and torques commands and motors/servo input are mapped between $[-1, -1]$
 - positive thrust and negative motor velocity are not allowed
- A custom control action can be implemented, but the commands must be properly converted to be used in PX4
- Parameter to compute the max wrench vectors are known from UAV sdf and used to extract normalized commands

- Omnidirectional tilting drone control using customized PX4



- An Overleaf template for the report can be found [here](#)
- A useful MATLAB example on how to present good plots can be found [here](#)
 - Output:



Thank you for your attention!

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